

Chapter 6

Probability Distribution

3-1 Using Statistics

Consider the different possible orderings of boy (B) and girl (G) in four sequential births. There are $2*2*2*2=2^4 = 16$ possibilities, so the sample space is:

BBBB	BGBB	GBBB	GGBB
BBBG	BGBG	GBBG	GGBG
BBGB	BGGB	GBGB	GGGB
BBGG	BGGG	GBGG	GGGG

If girl and boy are each equally likely [$P(G) = P(B) = 1/2$], and the sex of each child is independent of that of the previous child, then **the probability of each of these 16 possibilities is:**
 $(1/2)(1/2)(1/2)(1/2) = 1/16$.

Random Variables

Now count the **number of girls** in each set of four sequential births:

BBBB (0)	BGBB (1)	GBBB (1)	GGBB (2)
BBBG (1)	BGBG (2)	GBBG (2)	GGBG (3)
BBGB (1)	BGGB (2)	GBGB (2)	GGGB (3)
BBGG (2)	BGGG (3)	GBGG (3)	GGGG (4)

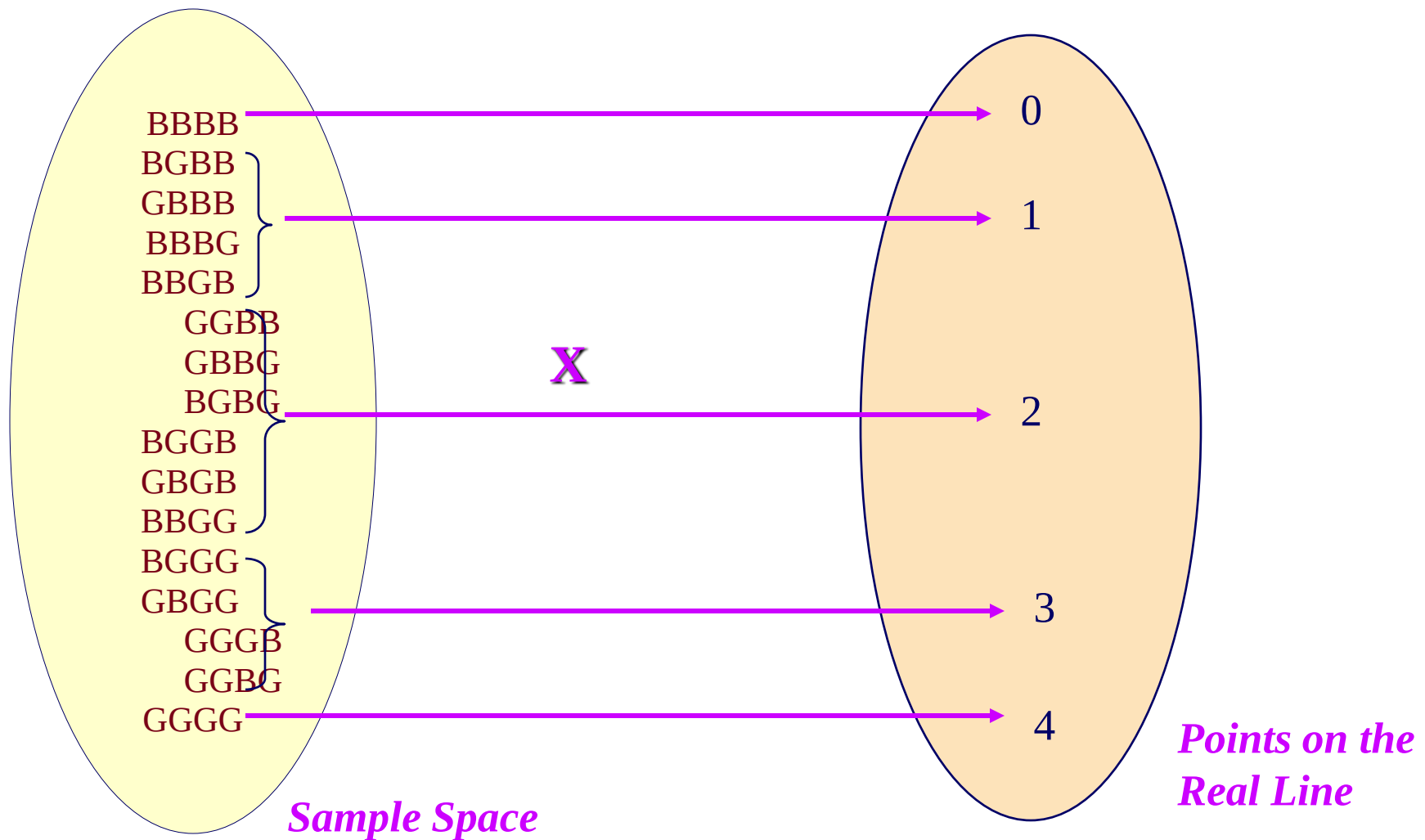
Notice that:

- each possible outcome is assigned a single numeric value,
- all outcomes are assigned a numeric value, and
- the value assigned varies over the outcomes.

The count of the number of girls is a **random variable**:

A **random variable, X** , is a function that assigns a single, but variable, value to each element of a sample space.

Random Variables (Continued)



Random Variables (Continued)

Since the random variable $X = 3$ when any of the four outcomes BGGG, GBGG, GGBG, or GGGB occurs,

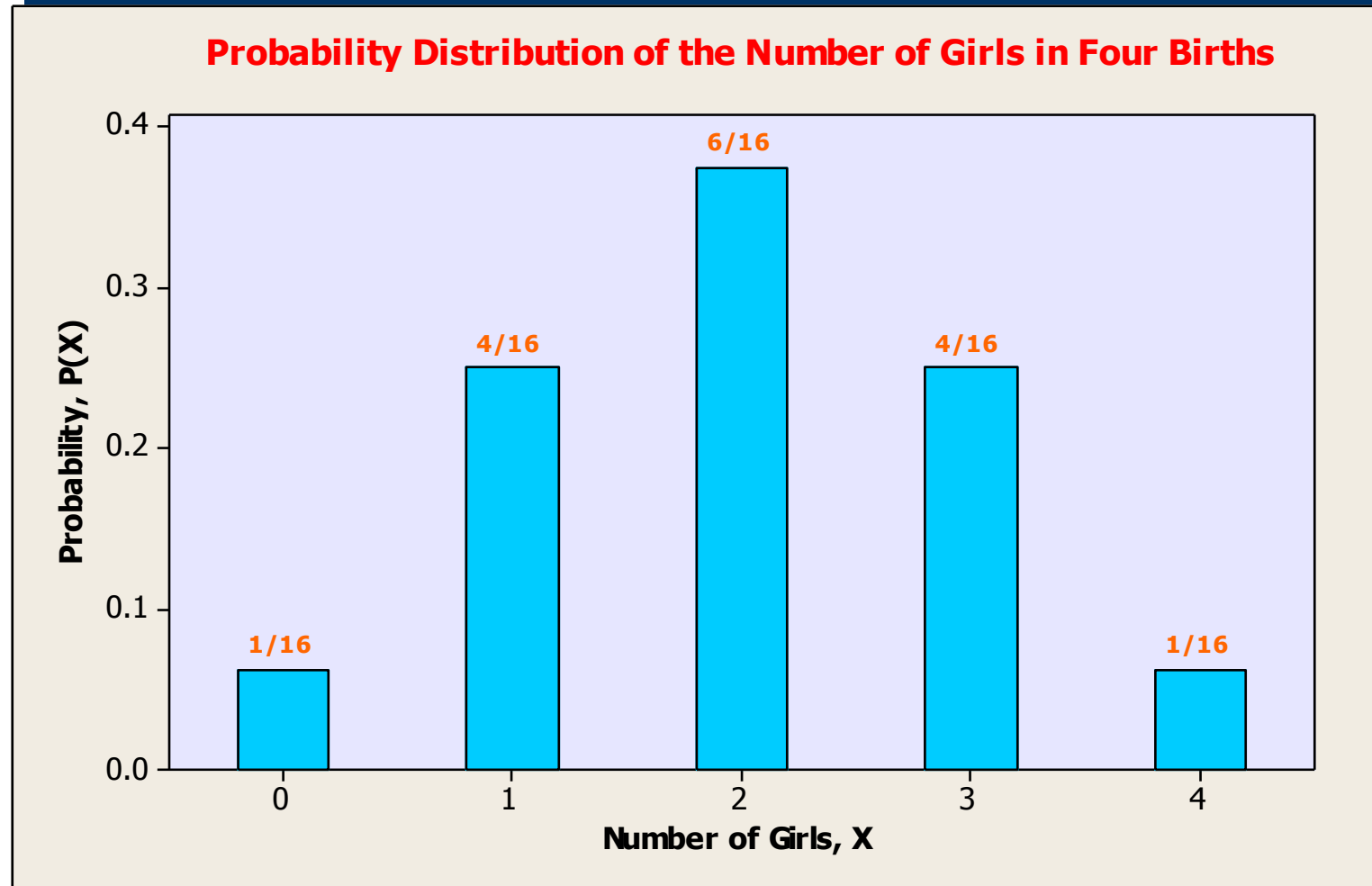
$$P(X = 3) = P(BGGG) + P(GBGG) + P(GGBG) + P(GGGB) = 4/16$$

The **probability distribution** of a random variable is a table that lists the possible values of the random variables and their associated probabilities.

<u>x</u>	<u>P(x)</u>
0	1/16
1	4/16
2	6/16
3	4/16
4	<u>1/16</u>
	16/16=1

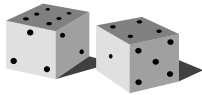
The Graphical Display for this Probability Distribution is shown on the next Slide.

Random Variables (Continued)



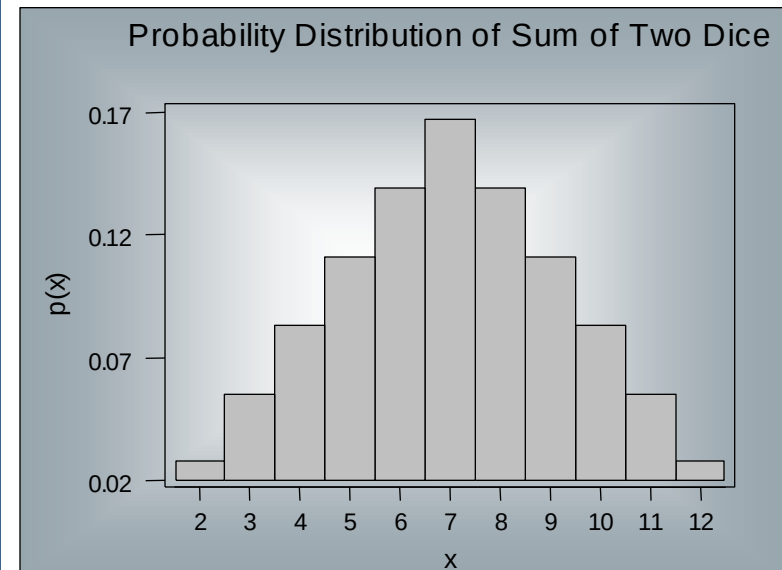
Example 3-1

Consider the experiment of **tossing two six-sided dice**. There are **36 possible outcomes**. Let the random variable **X** represent the **sum of the numbers on the two dice**:



	2	3	4	5	6	7
1,1	1,2	1,3	1,4	1,5	1,6	8
2,1	2,2	2,3	2,4	2,5	2,6	9
3,1	3,2	3,3	3,4	3,5	3,6	10
4,1	4,2	4,3	4,4	4,5	4,6	11
5,1	5,2	5,3	5,4	5,5	5,6	12
6,1	6,2	6,3	6,4	6,5	6,6	

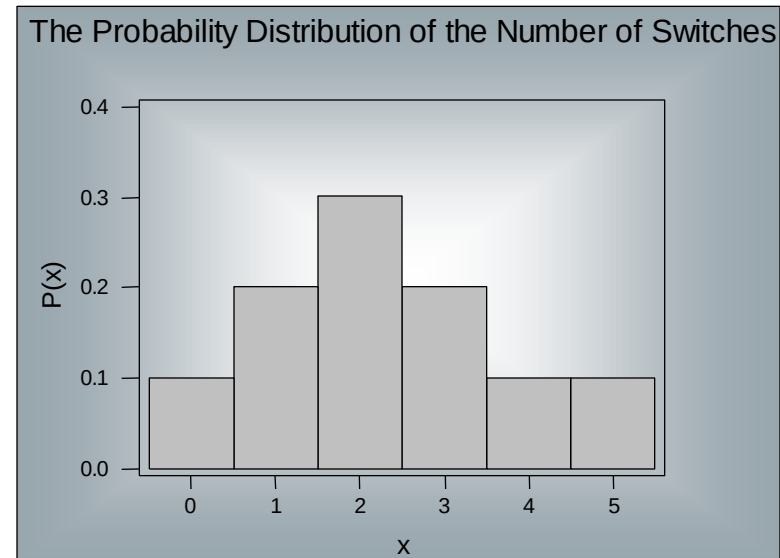
x	P(x)*
2	1/36
3	2/36
4	3/36
5	4/36
6	5/36
7	6/36
8	5/36
9	4/36
10	3/36
11	2/36
12	1/36
	1



Example 3-2

Probability Distribution of the Number of Switches

<u>x</u>	<u>P(x)</u>
0	0.1
1	0.2
2	0.3
3	0.2
4	0.1
5	<u>0.1</u>
	1



Probability of more than 2 switches:

$$P(X > 2) = P(3) + P(4) + P(5) = 0.2 + 0.1 + 0.1 = 0.4$$

Probability of at least 1 switch:

$$P(X \geq 1) = 1 - P(0) = 1 - 0.1 = .9$$

Discrete and Continuous Random Variables

A **discrete random variable**:

- has a countable number of possible values
- has discrete jumps (or gaps) between successive values
- has measurable probability associated with individual values
- counts

A **continuous random variable**:

- has an uncountably infinite number of possible values
- moves continuously from value to value
- has no measurable probability associated with each value
- measures (e.g.: height, weight, speed, value, duration, length)

Discrete Probability Distributions

The probability distribution of a discrete random variable X **must** satisfy the following two conditions.

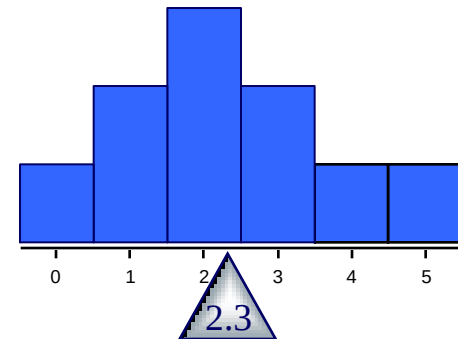
1. $P(x) \geq 0$ for all values of x .

$$2. \sum_{\text{all } x} P(x) = 1$$

[Corollary: $0 \leq P(X) \leq 1$]

3-2 Expected Values of Discrete Random Variables

The mean of a probability distribution is a **measure of its centrality or location**, as is the **mean or average of a frequency distribution**. It is a **weighted average**, with the values of the random variable weighted by their probabilities.



The mean is also known as the **expected value** (or **expectation**) of a random variable, because it is the value that is expected to occur, on average.

The **expected value** of a discrete random variable X is equal to the sum of each value of the random variable **multiplied** by its probability.

$$\mu = E(X) = \sum_{\text{all } x} xP(x)$$

x	$P(x)$	$xP(x)$
0	0.1	0.0
1	0.2	0.2
2	0.3	0.6
3	0.2	0.6
4	0.1	0.4
5	0.1	0.5
	1.0	2.3 = $E(X) = \mu$

Variance and Standard Deviation of a Random Variable

The **variance** of a random variable is the expectation of squared deviations from the mean:

$$\begin{aligned}\sigma^2 &= V(X) = E[(X - \mu)^2] = \sum_{all\ x} (x - \mu)^2 P(x) \\ &= E(X^2) - [E(X)]^2 = \left[\sum_{all\ x} x^2 P(x) \right] - \left[\sum_{all\ x} x P(x) \right]^2\end{aligned}$$

The **standard deviation** of a random variable is the square root of its variance: $\sigma = SD(X) = \sqrt{V(X)}$

Variance and Standard Deviation of a Random Variable – using Example 3-2

Table 3-8

Number of Switches, x	P(x)	xP(x)	(x-μ)	(x-μ) ²	P(x-μ) ²	<u>x²P(x)</u>
0	0.1	0.0	-2.3	5.29	0.529	0.0
1	0.2	0.2	-1.3	1.69	0.338	0.2
2	0.3	0.6	-0.3	0.09	0.027	1.2
3	0.2	0.6	0.7	0.49	0.098	1.8
4	0.1	0.4	1.7	2.89	0.289	1.6
5	0.1	<u>0.5</u>	2.7	7.29	<u>0.729</u>	<u>2.5</u>
		2.3			2.010	7.3

Recall: $\mu = 2.3$.

$$\sigma^2 = V(X) = E[(X - \mu)^2]$$

$$= \sum_{all\ x} (x - \mu)^2 P(x) = 2.01$$

$$= E(X^2) - [E(X)]^2$$

$$= \left[\sum_{all\ x} x^2 P(x) \right] - \left[\sum_{all\ x} x P(x) \right]^2$$

$$= 7.3 - 2.3^2 = 2.01$$

3-3 Bernoulli Random Variable

- If an experiment consists of a single trial and the outcome of the trial can only be either a success* or a failure, then the trial is called a **Bernoulli trial**.
- The number of success X in one Bernoulli trial, which can be 1 or 0, is a **Bernoulli random variable**.
- **Note:** If p is the probability of success in a Bernoulli experiment, the $E(X) = p$ and $V(X) = p(1 - p)$.

* The terms *success* and *failure* are simply statistical terms, and do not have positive or negative implications. In a production setting, finding a defective product may be termed a “success,” although it is not a positive result.

3-4 The Binomial Random Variable

Consider a **Bernoulli Process** in which we have a sequence of n identical trials satisfying the following conditions:

1. Each trial has two possible outcomes, called **success** *and **failure**. The two outcomes are **mutually exclusive** and **exhaustive**.
2. The **probability of success**, denoted by p , remains **constant** from trial to trial. The **probability of failure** is denoted by q , where $q = 1-p$.
3. The n trials are **independent**. That is, the outcome of any trial does not affect the outcomes of the other trials.

A random variable, X , that counts the number of successes in n Bernoulli trials, where p is the probability of success* in any given trial, is said to follow the **binomial probability distribution** with parameters n (number of trials) and p (probability of success). We call X the **binomial random variable**.

The Binomial Probability Distribution

The **binomial probability distribution**:

$$P(x) = \binom{n}{x} p^x q^{(n-x)} = \frac{n!}{x!(n-x)!} p^x q^{(n-x)}$$

where :

p is the probability of success in a single trial,

$q = 1-p$,

n is the number of trials, and

x is the number of successes.

Number of
successes, x

Probability $P(x)$

$$0 \quad \frac{n!}{0!(n-0)!} p^0 q^{(n-0)}$$

$$1 \quad \frac{n!}{1!(n-1)!} p^1 q^{(n-1)}$$

$$2 \quad \frac{n!}{2!(n-2)!} p^2 q^{(n-2)}$$

$$3 \quad \frac{n!}{3!(n-3)!} p^3 q^{(n-3)}$$

$\vdots$

$$n \quad \frac{n!}{n!(n-n)!} p^n q^{(n-n)}$$

1.00

Calculating Binomial Probabilities - Example

60% of products produced by a factory are defective. A random sample of 15 products are chosen. What is the probability that ***at most three*** of them will be defective?

n=15			
	p		
	.50	.60	.70
0	.000	.000	.000
1	.000	.000	.000
2	.004	.000	.000
3	.018	.002	.000
4	.059	.009	.001
...	...	...	...

Mean, Variance, and Standard Deviation of the Binomial Distribution

Mean of a binomial distribution:

$$\mu = E(X) = np$$

Variance of a binomial distribution:

$$\sigma^2 = V(X) = npq$$

Standard deviation of a binomial distribution:

$$\sigma = SD(X) = \sqrt{npq}$$

For example, if H counts the number of heads in five tosses of a fair coin :

$$\mu_H = E(H) = (5)(.5) = 2.5$$

$$\sigma_H^2 = V(H) = (5)(.5)(.5) = 1.25$$

$$\sigma_H = SD(H) = \sqrt{1.25} = 1.118$$

3-5 The Poisson Distribution

The **Poisson probability distribution** is useful for determining the probability of a number of occurrences over a given period of time or within a given area or volume. That is, the Poisson random variable counts occurrences over a continuous interval of time or space.

Poisson Distribution:

$$P(x) = \frac{\mu^x e^{-\mu}}{x!} \quad \text{for } x = 1, 2, 3, \dots$$

where μ is the mean of the distribution (which also happens to be the variance) and e is the base of natural logarithms ($e=2.71828\dots$).

The Poisson Distribution - Example

Example

A researcher found that the average occurrence of earthquake in North Africa within five years is six. Calculate the probability that in the coming five years there will be:

- a) no earthquake occur; b) exactly three earthquake occur;
- c) less than 2 earthquake occur; and d) at least one earthquake occur.

3-6 Continuous Random Variables

A **continuous random variable** is a random variable that can take on any value in an interval of numbers.

The probabilities associated with a continuous random variable X are determined by the **probability density function** of the random variable. The function, denoted $f(x)$, has the following properties.

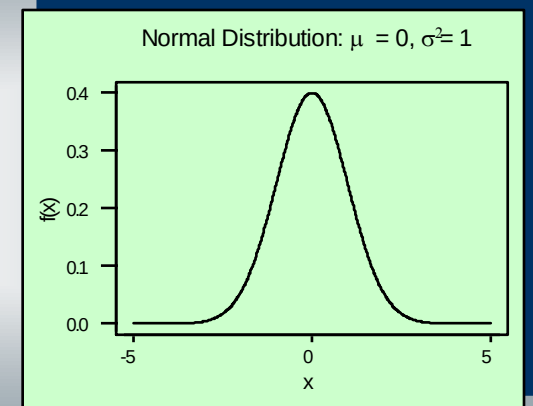
1. $f(x) \geq 0$ for all x .
2. The probability that X will be between two numbers a and b is equal to the area under $f(x)$ between a and b .
3. The total area under the curve of $f(x)$ is equal to 1.00.

The Normal Probability Distribution

The normal probability density function:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \text{for} \quad -\infty < x < \infty$$

where $e = 2.7182818 \dots$ and $\pi = 3.14159265 \dots$



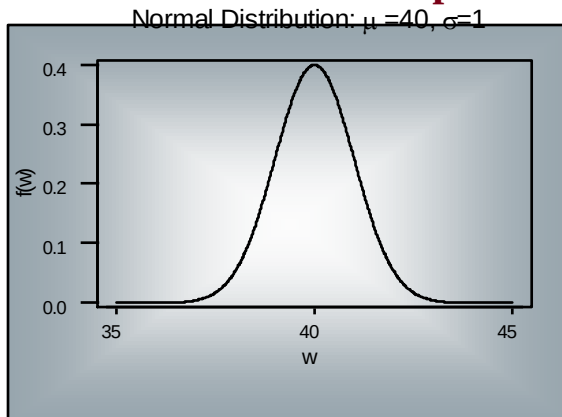
4-2 Properties of the Normal Distribution

- **The normal distribution**

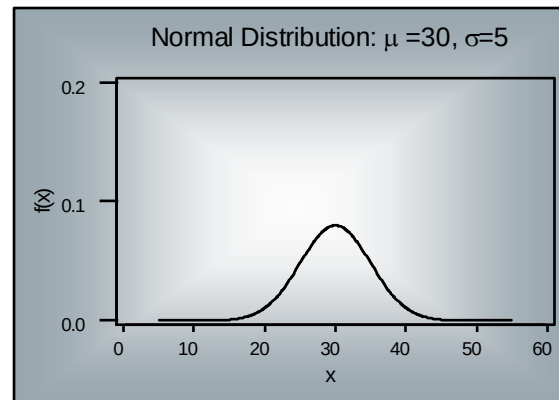
- ✓ Its curve is *bell-shaped, symmetric, and meso-kurtic* distributions.
- ✓ It *mean is μ* , and *variance is σ^2* . That is: $[X \sim N(\mu, \sigma^2)]$
- ✓ Its curve is *asymptotic* to the horizontal axis.
- ✓ The area under the normal probability density function is 1.

Normal Probability Distributions

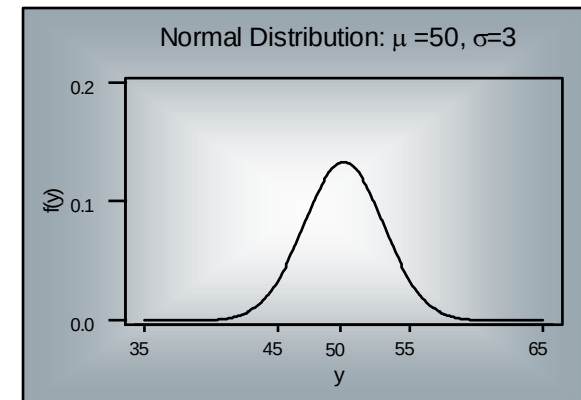
All of these are normal probability density functions, though each has a different mean and variance.



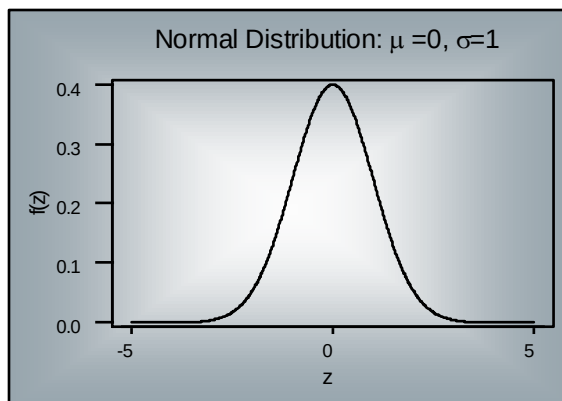
$$W \sim N(40, 1)$$



$$X \sim N(30, 25)$$



$$Y \sim N(50, 9)$$



$$Z \sim N(0, 1)$$

Consider:

$$P(39 \leq W \leq 41)$$

$$P(25 \leq X \leq 35)$$

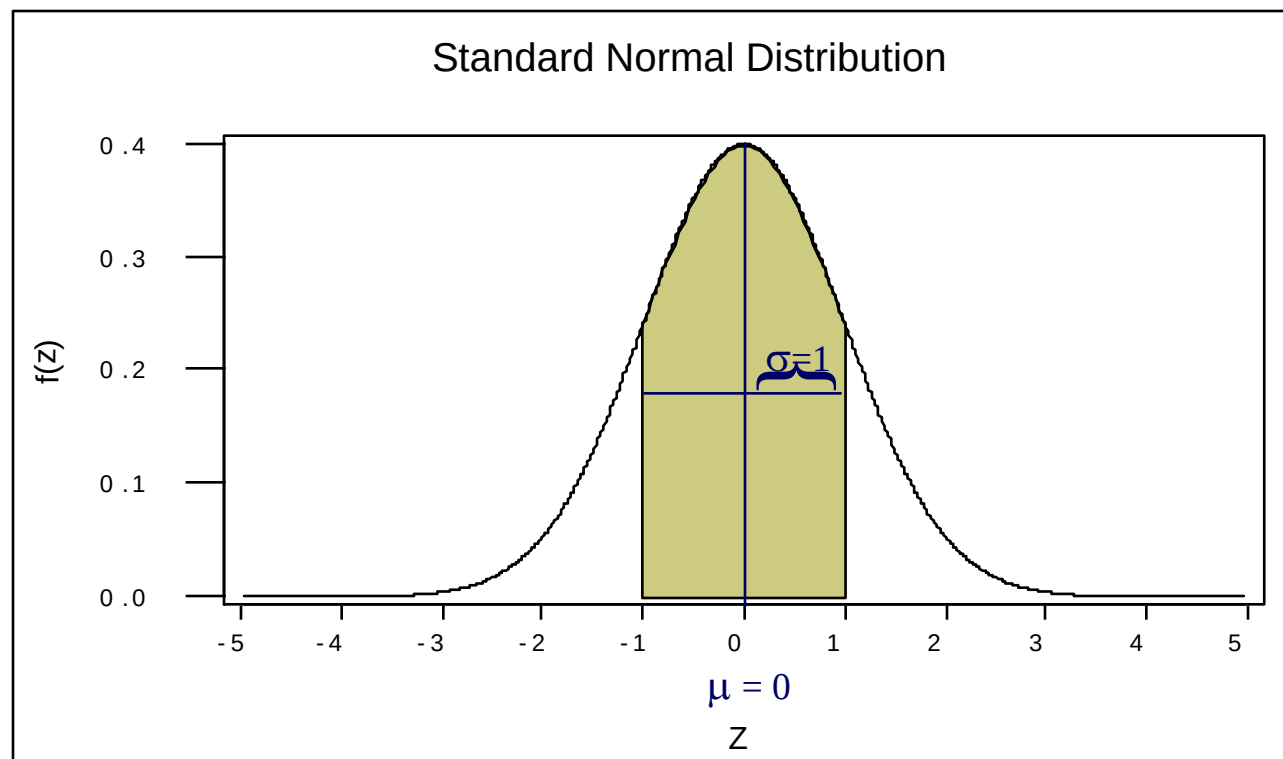
$$P(47 \leq Y \leq 53)$$

$$P(-1 \leq Z \leq 1)$$

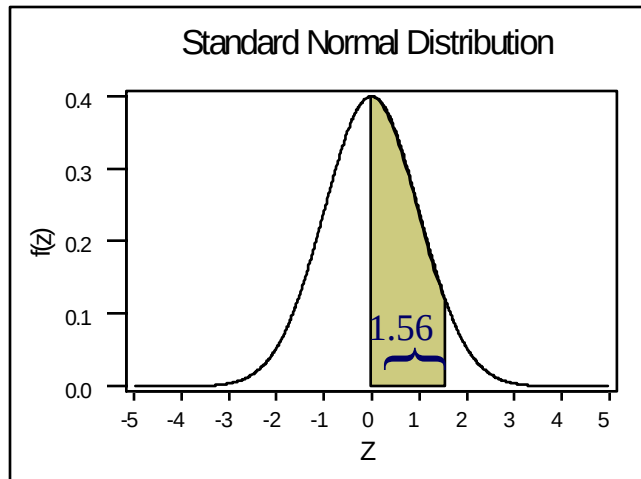
The probability in each case is an area under a normal probability density function.

4-4 The Standard Normal Distribution

The **standard normal random variable**, Z , is the normal random variable with mean $\mu = 0$ and standard deviation $\sigma = 1$: $Z \sim N(0, 1^2)$.



Finding Probabilities of the Standard Normal Distribution: $P(0 < Z < 1.56)$



Look in row labeled **1.5**
and column labeled **.06** to
find $P(0 \leq z \leq 1.56) =$
0.4406

Standard Normal Probabilities

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990

4-5 The Transformation of Normal Random Variables

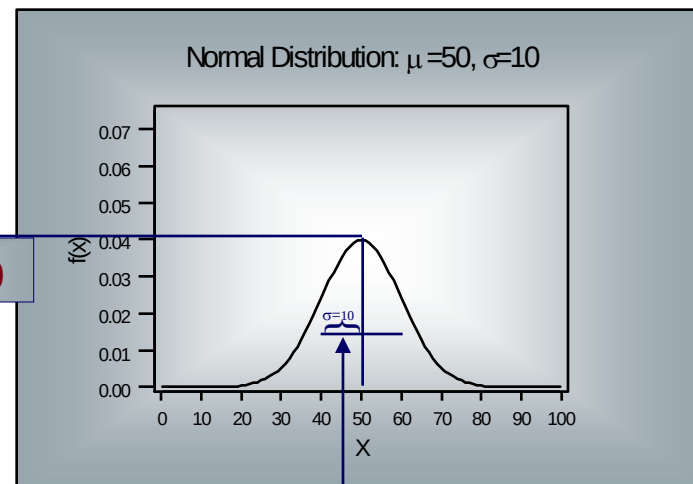
The area within $k\sigma$ of the mean is the same for all normal random variables. So an area under any normal distribution is equivalent to an area under the standard normal. In this example: $P(40 \leq X \leq 60) = P(-1 \leq Z \leq 1) = 0.6826$ since $\mu = 50$ and $\sigma = 10$.

The transformation of X to Z :

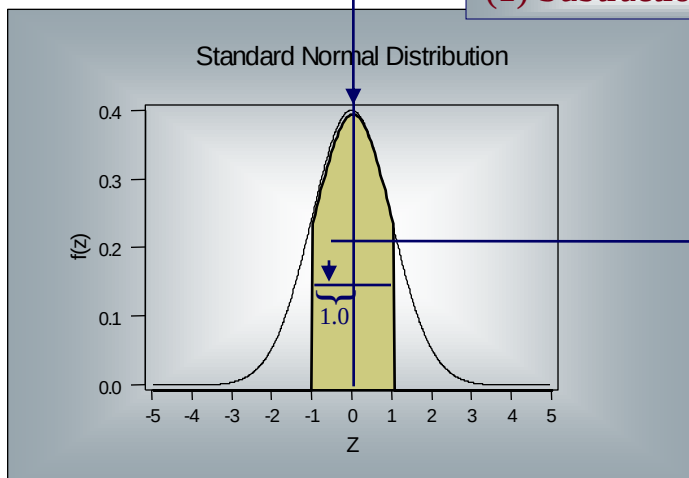
$$Z = \frac{X - \mu_x}{\sigma_x}$$

Transformation

(1) Subtraction: $(X - \mu_x)$



(2) Division by σ_x



The inverse transformation of Z to X :

$$X = \mu_x + Z \sigma_x$$

Using the Normal Transformation

Example 4-9

$$X \sim N(160, 30^2)$$

$$\begin{aligned} &P(100 \leq X \leq 180) \\ &= P\left(\frac{100 - \mu}{\sigma} \leq \frac{X - \mu}{\sigma} \leq \frac{180 - \mu}{\sigma}\right) \\ &= P\left(\frac{100 - 160}{30} \leq Z \leq \frac{180 - 160}{30}\right) \\ &= P(-2 \leq Z \leq .6667) \\ &= 0.4772 + 0.2475 = 0.7247 \end{aligned}$$

Example 4-10

$$X \sim N(127, 22^2)$$

$$\begin{aligned} &P(X < 150) \\ &= P\left(\frac{X - \mu}{\sigma} < \frac{150 - \mu}{\sigma}\right) \\ &= P\left(Z < \frac{150 - 127}{22}\right) \\ &= P(Z < 1.045) \\ &= 0.5 + 0.3520 = 0.8520 \end{aligned}$$

The t distribution

It is useful in inference concerning mean

The distribution has one parameter, degrees of freedom

It has similar properties as normal distribution,